Yuto Kuwayama (桑山 裕斗)

Curriculum Vitae

Personal information

Date of birth	12 August 2001
Nationality	Japan
Affiliation	Theoretical Astrophysics, Department of Earth and Space Science, The University of Osaka, 1-1 Machikaneyama, Toyonaka, Osaka 560-0043, Japan
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Research experience and Education

2025– **Master course of Science**, *The University of Osaka*, Japan.
Supervisor: Prof. Kentaro Nagamine

2021–2025 **Bachelor of Science (Physics)**, *The University of Osaka*, Japan.
Supervisor: Prof. Kentaro Nagamine
Thesis title: Exploring the Density Field that Forms a Quasar Overdensity
(クェーサー高密度領域を形成する密度場の探索)

Publications

Leading Author Publications

2026 **Yuto Kuwayama, Yongming Liang, Kentaro Nagamine, Yuri Oku, Daisuke Nishihama, Daisuke Toyouchi, Keita Fukushima, Hidenobu Yajima, Hyunbae Park, Masami Ouchi**.
Cosmic Himalayas in CROCODILE : Probing the Extreme Quasar Overdensities by Count-in-Cells analysis and Nearest Neighbor Distribution
The Astrophysical Journal, **1001**, 172 (2026)
DOI: 10.3847/1538-4357/ae5798 arXiv: 2512.24966
(Submitted Dec. 2025, accepted Apr. 2026)

Scientific talks & Presentation

Contributed talks

Sep. 2026 **Galaxy-IGM Workshop 2026**, *Hotel Sun Valley Nasu, Tochigi*, Japan.
Radiative Transfer of Hα and Lyα Lines: Investigating the Environmental Dependence of Line Propagation

Sep. 2026 **Astronomical Society of Japan Autumn Annual Meeting 2026 (**日本天文学会 **2026**年秋季年会**)**, *Okayama University of Science, Okayama*, Japan.
Cosmic Himalayas in CROCODILE: Investigation of quasar overdensities at Cosmic Noon by Cosmological hydrodynamic simulation and Mock Observation

Apr. 2026 **CAMELS + Baryon Pasting Workshop 2026**, *Nakanoshima Center, The University of Osaka*, Japan.
Cosmic Himalayas in CROCODILE: Why Extreme Overdensities are observed? Statistical investigations of extreme overdensities in simulation

Sep. 2025 **Astronomical Society of Japan Autumn Annual Meeting 2025 (日本天文学会 2025年秋季年会)**, *Kaikyo Messe Shimonoseki, Yamaguchi*, Japan.
Exploring Extreme Quasar Overdensities at $z \sim 2$ by cosmological hydrodynamic simulation
Does ΛCDM model produce the Extreme Quasar Overdensity like "Cosmic Himalayas"?

July 2025 **Galaxy-IGM Workshop 2025**, *Hotel Sun Valley Nasu, Tochigi*, Japan.
How rare are the Extreme Overdensity of Quasars? Insight from 1-pt and 2-pt statistics Using CROCODILE simulation

May 2025 **International Workshop on Galaxy Formation + AGORA in Asia 2025**, *Nakanoshima Center, the University of Osaka*, Japan.
How Rare Are the Cosmic Himalayas? Insights from Cosmological Simulations

Mar. 2025 **SPR+ILR joint workshop**, *Mitaka campus, The University of Tokyo*, Japan.
Exploring the Density Field that forms a Quasar Overdensity at $z \sim 2$

Mar. 2025 **Elucidating the Material Circulation in the Early Universe**, *Komaba campus, The University of Tokyo*, Japan.
Exploring the Density Field that Forms a Quasar Overdensity at $z \sim 2$

Posters

Aug. 2026 **The 12th Galaxy Evolution Workshop**, *Seoul National University, Seoul*, Korea.
A Comprehensive Study of an Overdensity at Cosmic Noon: Distribution of Hα Emitters and Lyα Emitters
(Poster + Flash Talk)

May 2026 **KEK Cosmo 2026**, *KEK, Ibaraki*, Japan.
Statistical Evaluation of High-Density Quasar Environments via One- and Two-Point Statistics in Cosmological Hydrodynamic Simulations

July 2025 **55th astronomical meeting for young researchers**, *Chikuma, Nagano*, Japan.
How rare are the Extreme Overdensity of Quasars? Insight from 1-pt and 2-pt statistics Using CROCODILE simulation

School

Mar. 2026 **NAOJ CfCA Hydrodynamics Shool (国立天文台CfCA流体学校)**, *National Astronomical Observatory of Japan*, Mitaka, Tokyo, Japan.

Jun. 2025 **YITP International School - East Asian Meeting on Large Galaxy Surveys for Cosmology and Galaxy Formation**, *Masukawa Hall, Yukawa Institute for Theoretical Physics, Kyoto University (YITP)*, Japan.

Feb. 2025 **Spring school of N-body simulation (N体シミュレーション立春の学校)**, *National Astronomical Observatory of Japan*, Mitaka, Tokyo, Japan.

Languages

Japanese	Native
English	Upper Intermediate
German	Basic